

1. Introduction to Structural Biology

Definition

Structural biology is a branch of molecular biology, biochemistry, and biophysics concerned with the **molecular structure of biological macromolecules**—primarily proteins and nucleic acids—and how alterations in their structures affect their function.

Explanation

It investigates how biological molecules acquire the structures they have, how structural changes affect their functions, and how these structures interact within cells. It uses various experimental methods such as:

- X-ray crystallography
- NMR spectroscopy
- Cryo-electron microscopy
- Computational modeling

Diagram

DNA → mRNA → Protein → 3D Structure → Function

Applications

- **Drug Design:** Structural knowledge allows pharmaceutical companies to design drugs that precisely target specific proteins.
- **Disease Understanding:** Helps in understanding how mutations cause diseases (e.g., cancer, genetic disorders).
- **Biotechnology:** Engineered enzymes and proteins.

Examples

- Hemoglobin structure explains oxygen transport.

- 3D structure of the SARS-CoV-2 spike protein guided vaccine development.
-

2. Macromolecules

Definition

Macromolecules are **large, complex molecules** essential for life, made by polymerization of smaller subunits. The four major types are:

1. **Proteins**
2. **Nucleic acids (DNA & RNA)**
3. **Carbohydrates**
4. **Lipids**

Explanation

Each macromolecule performs different functions:

- **Proteins:** Catalysts (enzymes), structural roles.
- **Nucleic Acids:** Genetic information storage (DNA) and transfer (RNA).
- **Carbohydrates:** Energy storage and structural.
- **Lipids:** Membrane formation and energy storage.

Examples

- **Proteins:** Insulin, hemoglobin
- **Nucleic Acids:** DNA in chromosomes
- **Carbohydrates:** Starch, cellulose
- **Lipids:** Cholesterol, phospholipids

Applications

- **Medical field:** Understanding disease mechanisms.
- **Genetic engineering:** Recombinant DNA technology.
- **Agriculture:** GMOs, improving crop yield.

Diagram

Monomers → Polymers

Amino acids → Proteins

Nucleotides → DNA/RNA

Monosaccharides → Polysaccharides

Fatty acids + Glycerol → Lipids

3. Sequence–Structure–Function Relationship, Screening Target Proteins and Its Structure

Definition

The **sequence–structure–function relationship** refers to how a **protein's amino acid sequence (primary structure)** determines its **3D structure**, which in turn defines its **biological function**.

Explanation

- **Primary structure** → sequence of amino acids
- **Secondary structure** → alpha helices & beta sheets
- **Tertiary structure** → 3D folding
- **Quaternary structure** → protein complexes

Target protein screening refers to identifying proteins that are involved in a disease or biological process and can be used as a drug target.

Diagram

Sequence (Amino acids)



Structure (3D folding)



Function (Biological role)

Examples

- Mutation in CFTR protein → misfolding → cystic fibrosis
- HIV protease structure used in designing protease inhibitors

Applications

- **Drug Discovery:** Target proteins are analyzed for docking studies.
- **Personalized Medicine:** Based on protein mutations in individuals.
- **Enzyme Engineering:** Altering active sites for new functions.

Protein Structure Screening Methods

1. **X-ray crystallography** – detailed static structure
2. **Cryo-EM** – large complexes at near-atomic resolution
3. **Homology modeling** – computational prediction
4. **Molecular Docking** – for screening ligands

